DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Examination – Summer 2022

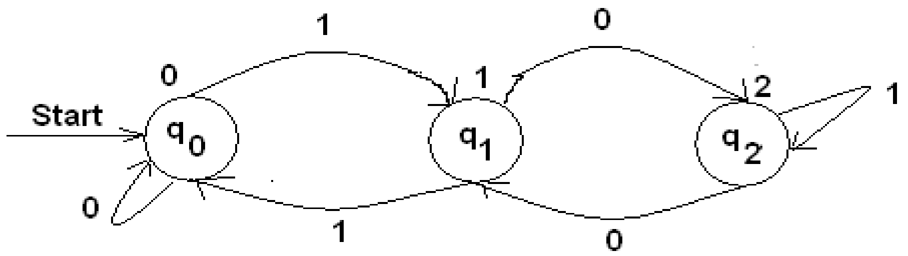
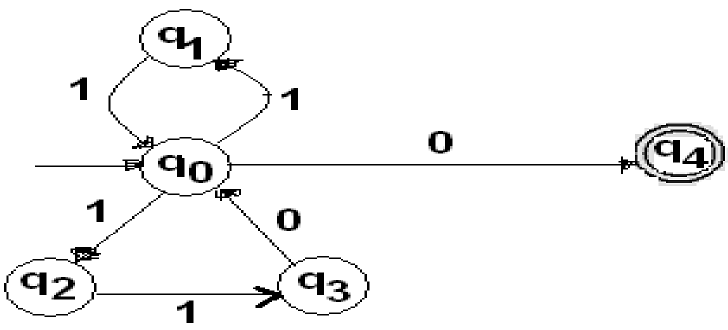
Course: B. Tech. **Branch : CSE** **Semester :5th**

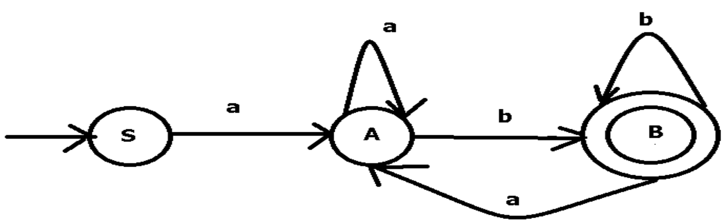
Subject Code & Name: BTCOC502: Theory of Computations

Max Marks: 60 **Date:** **Duration: 3 Hr.**

Instructions to the Students:

1. *All the questions are compulsory.*
2. *The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.*
3. *Use of non-programmable scientific calculators is allowed.*
4. *Assume suitable data wherever necessary and mention it clearly.*

		(Level/CO)	Marks
Q. 1	**Solve Any Two of the following.**		**2*6= 12M**
A)	**Find DFA that accepts only the language of all strings with b as the second letter over the alphabet Σ={a,b}.**		**6M**
B)	**Construct a Mealy machine of the following Moore machine**  		**6M**
C)	**Construct the DFA for the following NFA**  		**6M**
Q.2	**Solve Any Two of the following.**		**2*6= 12M**
A)	**Explain Chomsky classification of grammars**		**6M**
B)	**Consider the grammar given as G = ({S,A}, {a,b}, P , S)** **Where production P consists of ,**		**6M**

	S→ aAS | a A→ SbA | SS | ba Find a) Leftmost derivation b) Rightmost derivation c) Parse tree for the string " aabbaa ".		
C)	Find a reduced grammar G to the grammar given below S→ AB|CA A→ a B→BC|AB C→aB|b		6M
Q. 3	Solve Any One of the following.		2*6= 12M
A)	Convert the following CFG to CNF: S→ aSa| bSb | a | b | aa | bb		6M
B)	Obtain RLG from the given Finite Automata  		6M
C)	For the grammar given find L(G) S → a | Sa | b | bS .		6M
Q.4	Solve Any Two of the following.		2*6= 12M
A)	Construct a PDA accepting L = { $a^n b^m a^n$ |m, n ≥1} by a null stack. Is it deterministic?		6M
B)	Construct a PDA equivalent to the following context free grammar S→aAA A→aS|bS|a		6M

C)	**Let M = ({ q_0 , q_1} , { 0 , 1}, { X , Z_0} , δ , q_0 , Z_0 , Φ}** **where δ is given by** $\delta(q_0, 0, Z_0) = \{ (q_0, X Z_0) \}$ $\delta(q_0, 0, X) = \{ (q_0, X X) \}$ $\delta(q_0, 1, X) = \{ (q_1, \varepsilon) \}$ $\delta(q_1, 1, X) = \{ (q_1, \varepsilon) \}$ $\delta(q_1, \varepsilon, X) = \{ (q_1, \varepsilon) \}$ $\delta(q_1, \varepsilon, Z_0) = \{ (q_1, \varepsilon) \}$ **Construct a CFG**		6M
Q. 5	**Solve Any One of the following.**		**2*6= 12M**
A)	**Obtain a Turing machine to accept the language** **L= { $0^n 1^n 2^n$ | n≥ 1}.**		6M
B)	**Design a TM to find one's complement of the binary number**.		6M
C)	**Explain the following** **1)Turing Machine with stay-option** 2) **Multiple Tapes Turing Machine**		6M
	*** End ***		

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